LongE2V: Long-Horizon Event-based Video Reconstruction, Prediction, and Frame Interpolation with Video Diffusion Models

CHENG-DE FAN, National Yang Ming Chiao Tung University, Taiwan
CHUN-WEI TUAN MU, National Yang Ming Chiao Tung University, Taiwan
CHEN-WEI CHANG, National Yang Ming Chiao Tung University, Taiwan
CHIN-YANG LIN, National Yang Ming Chiao Tung University, Taiwan
KUN-RU WU, National Yang Ming Chiao Tung University, Taiwan
YU-CHEE TSENG, National Yang Ming Chiao Tung University, Taiwan
YU-LUN LIU, National Yang Ming Chiao Tung University, Taiwan

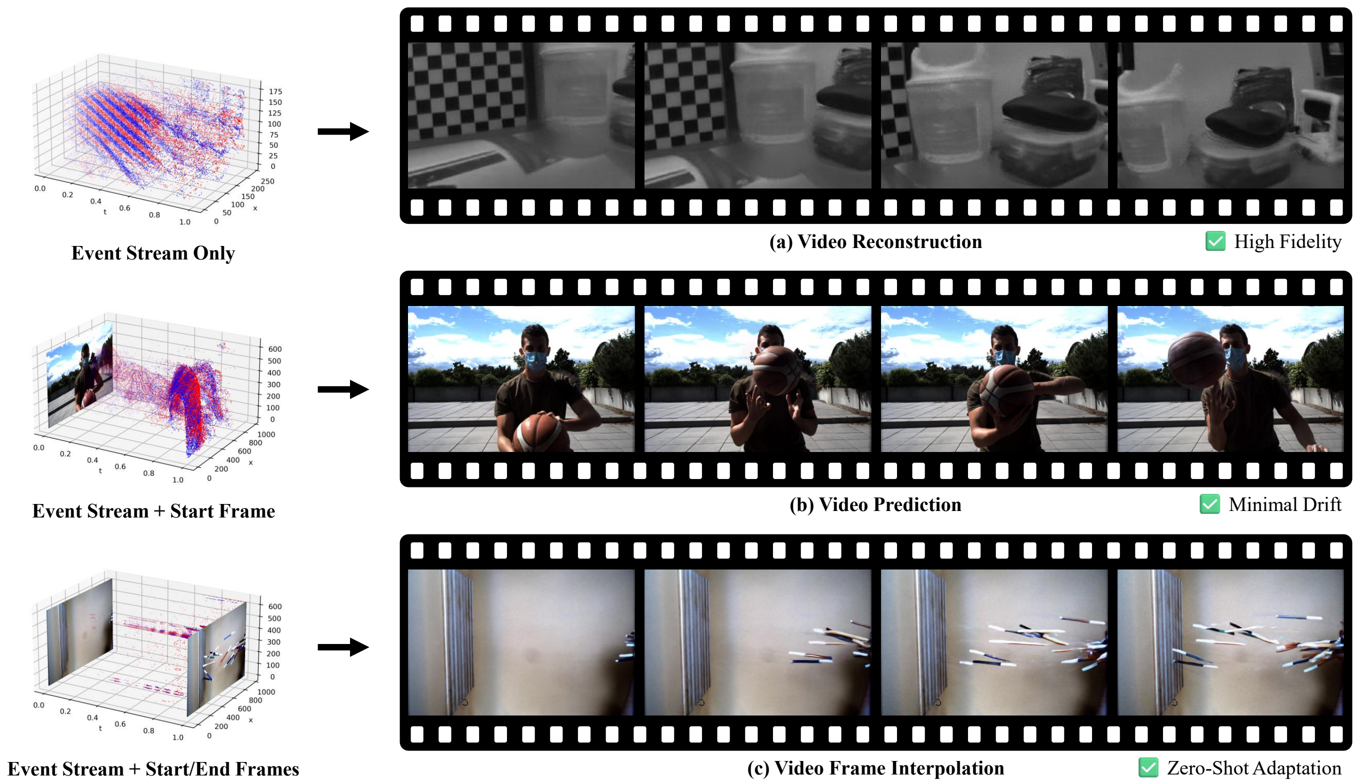

Fig. 1. **Event-based video generation.** We leverage pre-trained video diffusion priors to address three distinct inverse problems within a single architecture. Depending on the input condition, our model performs: (a) **Video Reconstruction**, recovering high-fidelity textures from sparse event streams, (b) **Video Prediction**, generating long-term sequences from a single start frame with minimal drift via our autoregressive unrolling strategy, and (c) **Video Frame Interpolation**, achieving zero-shot adaptation to synthesize intermediate frames by leveraging event dynamics as temporal guidance.

Recovering high-quality video from sparse event streams is a challenging task. Regression methods often blur textures, while existing generative models struggle with long-term stability. We propose LongE2V, a novel approach that leverages pre-trained video diffusion priors to jointly handle event-based video reconstruction, prediction, and frame interpolation. By fine-tuning a foundational video model, our approach achieves high data

Authors' Contact Information: Cheng-De Fan, National Yang Ming Chiao Tung University, Taiwan, fansam39@gmail.com; Chun-Wei Tuan Mu, National Yang Ming Chiao Tung University, Taiwan, raytm9999@gmail.com; Chen-Wei Chang, National Yang Ming Chiao Tung University, Taiwan, steven891213.ii12@nycu.edu.tw; Chin-Yang Lin, National Yang Ming Chiao Tung University, Taiwan, linjohn0903@gmail.com; Kun-Ru Wu, National Yang Ming Chiao Tung University, Taiwan, wufish@nycu.edu.tw; Yu-Chee Tseng, National Yang Ming Chiao Tung University, Taiwan, yctseng@nycu.edu.tw; Yu-Lun Liu, National Yang Ming Chiao Tung University, Taiwan, yulunliu@cs.nycu.edu.tw.

SIGGRAPH Conference Papers '26, Los Angeles, CA, USA

ACM ISBN 979-8-4007-2554-8/2026/07
https://doi.org/10.1145/3799902.3811151

efficiency and superior perceptual quality. We introduce Autoregressive Unrolling and Adaptive Context Switching to mitigate temporal drift in extremely long sequences. We also propose Reencoding Alignment with Cross Residual Correction to ensure precise bidirectional consistency during frame interpolation. Furthermore, Event Voxel Density Augmentation ensures robustness across varying sensor resolutions. Extensive experiments on real-world benchmarks demonstrate that LongE2V outperforms state-of-the-art methods across all three tasks, exhibiting exceptional temporal coherence and zero-shot generalization.

CCS Concepts: • **Computing methodologies** → **Reconstruction**; **Computational photography**; *Neural networks*; • **Hardware** → Sensors and actuators.

ACM Reference Format:
Cheng-De Fan, Chun-Wei Tuan Mu, Chen-Wei Chang, Chin-Yang Lin, Kun-Ru Wu, Yu-Chee Tseng, and Yu-Lun Liu. 2026. LongE2V: Long-Horizon Event-based Video Reconstruction, Prediction, and Frame Interpolation with Video Diffusion Models. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers (SIGGRAPH Conference Papers '26), July 19–23, 2026, Los Angeles, CA, USA.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3799902.3811151

1 Introduction

Event cameras are bio-inspired sensors capturing asynchronous brightness changes with microsecond resolution and high dynamic range (HDR). Unlike standard cameras prone to motion blur, they excel in high-speed dynamics. However, their sparse, intensity-free output is incompatible with standard vision algorithms, making high-fidelity video recovery an inherently ill-posed problem. We address this by leveraging video diffusion models for Video Reconstruction, Prediction, and Frame Interpolation. As illustrated in Fig. 1, a robust solution must simultaneously recover photometric details (Reconstruction), ensure long-term coherence (Prediction), and enable zero-shot intermediate frame generation (Frame Interpolation), effectively bridging neuromorphic sensing and human-interpretable vision.

Traditional event-based video generation relies on CNNs or RNNs [Rebecq et al. 2019b; Scheerlinck et al. 2020], with pioneers like E2VID [Rebecq et al. 2019b] and FireNet [Scheerlinck et al. 2020] aggregating temporal information. Recent advances employ Transformers [Weng et al. 2021] and Hypernetworks [Ercan et al. 2024] to enhance representation. With the rise of generative AI, Video Diffusion Models (VDMs)[Blattmann et al. 2023a; Ho et al. 2022b] like VDM-EVFI[Chen et al. 2025a] have been adapted for interpolation. These methods establish strong baselines by mapping event volumes to frames and are often trained on synthetic datasets.

Despite advancements, challenges persist in real-world scenarios. Regression-based methods like E2VID suffer from "regression-to-the-mean," yielding blurry textures (Fig.2(a)). While diffusion models improve generation, naive application causes instability; long-term prediction suffers from severe error accumulation and drift (Fig.2(b)). Furthermore, interpolation methods struggle with fast, complex dynamics, producing ghosting artifacts (Fig. 2(c)). Crucially, many existing methods are tailored to individual tasks, often requiring separate architectures for reconstruction, prediction, and frame interpolation, thus limiting flexibility.

To address these limitations, we propose **LongE2V**, leveraging pre-trained Video Diffusion Models (CogVideoX [Yang et al. 2024]) to handle reconstruction, prediction, and frame interpolation. As shown in Fig. 4, we formulate these tasks as conditional generation driven by event voxels. To ensure long-term stability, we introduce Autoregressive Unrolling combined with Adaptive Context Switching, which dynamically adjusts temporal dependencies to mitigate error accumulation and drift. For interpolation, we propose Reencoding Alignment and Cross Residual Correction to resolve temporal misalignments in the 3D VAE latent space. Finally, Event Voxel Density Augmentation ensures robustness across varying sensor resolutions.

Our contributions are summarized as follows:

- We propose LongE2V, leveraging pre-trained video diffusion priors to handle event-based reconstruction, prediction, and frame interpolation with superior data efficiency.
- We introduce Autoregressive Unrolling and Adaptive Context Switching to ensure long-term stability in reconstruction and prediction, alongside Reencoding Alignment with Cross Residual Correction to ensure temporal consistency in frame interpolation.
- We design Event Voxel Density Augmentation to achieve robust generalization across different sensor resolutions, demonstrating that our method outperforms SOTA baselines across all three tasks and achieves superior perceptual quality, stability, and zero-shot generalization.

2 Related Work

Event-based Video Reconstruction. Reconstructing intensity from relative brightness changes is ill-posed [Gallego et al. 2020]. Early optimization methods [Bardow et al. 2016; Munda et al. 2018; Scheerlinck et al. 2018; Zhang et al. 2022] were superseded by deep learning. E2VID [Rebecq et al. 2019a,b] established recurrent U-Net baselines trained on synthetic data [Rebecq et al. 2018], followed by improvements in efficiency [Cadena et al. 2023; Scheerlinck et al. 2020], sim-to-real transfer [Cadena et al. 2021; Ercan et al. 2024; Stoffregen et al. 2020], and architectures including Transformers [Weng et al. 2021], SSL [Paredes-Vallés and De Croon 2021; Wang et al. 2024b], SNNs [Zhu et al. 2020, 2022], and GANs [Wang et al. 2019]. Recent generative approaches employ language [Chen et al. 2024a], diffusion [Liang et al. 2024], or temporal residuals [Zhu et al. 2024]. In contrast, our method handles reconstruction, prediction, and frame interpolation via efficient fine-tuning.

Event-based Video Frame Interpolation. EVFI exploits high temporal resolution to synthesize intermediate frames. Approaches include warping-synthesis hybrids [Cho et al. 2024; Kim et al. 2023; Ma et al. 2024a; Sun et al. 2024, 2023; Tulyakov et al. 2022, 2021] and flow-based methods employing cycle-consistency [He et al. 2022] or adaptive computation [Liu et al. 2024b; Shi et al. 2023; Wu et al. 2022; Zhang et al. 2025b; Zhang and Yu 2022]. Recently, EPA [Liu et al. 2025] further leverages fine-grained event cues to guide hierarchical feature alignment within a perceptual space. Direct synthesis methods [Liu et al. 2024a; Paikin et al. 2021] have recently adapted pre-trained video diffusion models [Chen et al. 2025a]. Unlike prior

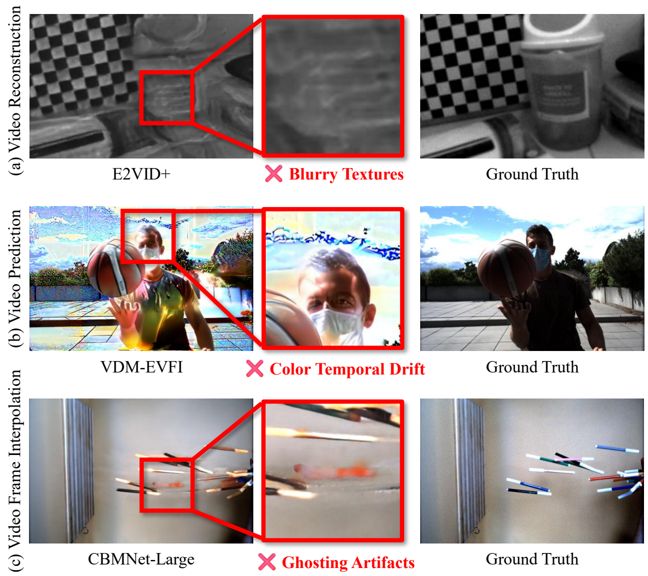

Fig. 2. **Challenges in event-based video generation.** We highlight failure cases in state-of-the-art methods: (a) Reconstruction: Regression-based methods (e.g., E2VID+[Stoffregen et al. 2020]) suffer from "regression-to-the-mean," causing **blurry textures** and detail loss. (b) Prediction: Direct video diffusion (e.g., VDM-EVFI[Chen et al. 2025a]) on long sequences suffers from error accumulation, leading to severe **color temporal drift**. (c) Interpolation: Existing networks (e.g., CBMNet-Large [Kim et al. 2023]) fail to capture complex intermediate motion, producing significant **ghosting artifacts**. Our method leverages video diffusion priors to ensure high-fidelity and stable generation across all tasks.

work restricted to interpolation, we extend diffusion priors to reconstruction and prediction, introducing *Reencoding Alignment* and *Cross Residual Correction* to resolve 3D VAE temporal misalignment.

Video Diffusion Models. Diffusion models [Ho et al. 2020; Song et al. 2020] evolved from U-Net architectures optimizing attention and efficiency [Bar-Tal et al. 2024; Blattmann et al. 2023a,b; Ho et al. 2022a,b; Singer et al. 2022] to Diffusion Transformers (DiTs) achieving state-of-the-art quality [Chen et al. 2024b; Ma et al. 2024c; Yang et al. 2024]. Foundation models now scale to minute-long generation [Brooks et al. 2024; Kong et al. 2024; Team 2024; Zheng et al. 2024] using flow matching [Chen et al. 2025b; Esser et al. 2024; Yin et al. 2025]. Building on CogVideoX [Yang et al. 2024], we utilize events for *explicit motion guidance*, unlike the implicit motion learning in text-to-video generation.

Controllable Video Generation. Control methods range from spatial conditioning via adapters or attention [Chen et al. 2023a; Guo et al. 2023; Mou et al. 2024; Wang et al. 2023; Zhang et al. 2023] to fine-grained trajectory control [Geng et al. 2025; He et al. 2024; Jin et al. 2025; Ma et al. 2024b; Namekata et al. 2024; Wang et al. 2024d,c; Wu et al. 2024; Zhang et al. 2025a]. Structure-guided approaches separate content from motion [Esser et al. 2023; Niu et al. 2024; Wang et al. 2024a; Xiao et al. 2024]. We posit that event streams offer ideal structural conditions, encoding natural scene dynamics with microsecond resolution without manual specification.

Long-term Video Generation. Long video generation combats error accumulation [Bengio et al. 2015; Lamb et al. 2016] via memory architectures [Henschel et al. 2025; Yin et al. 2023; Zhao et al. 2024] or training-free noise rescheduling [Chen et al. 2025c, 2023b; Kim et al. 2024; Lu et al. 2024; Qiu et al. 2023; Zhou et al. 2024]. Recent works target the train-inference gap directly [Guo et al. 2025; Huang et al. 2025; Yin et al. 2025; Yoo et al. 2023; Zhang and Agrawala 2025]. We introduce *Autoregressive Unrolling* to fine-tune on predictions [Bengio et al. 2015; Lamb et al. 2016] and *Adaptive Context Switch* to dynamically update context, preventing temporal drift unlike fixed-schedule methods.

3 Method

3.1 Preliminaries

Video Diffusion Models. We adopt CogVideoX I2V [Yang et al. 2024], which encodes video X into latents Z_0 using a 3D VAE with 4× temporal and 8× spatial compression. The Diffusion Transformer (DiT) ϵ_θ is trained to minimize the denoising objective: $\mathcal{L} = \mathbb{E}_{Z_0,t,C,\epsilon}\left[\|\epsilon - \epsilon_\theta(Z_t, t, C)\|_2^2\right]$, where Z_t are the noisy latents and C denotes conditioning signals. During inference, the model iteratively denoises random Gaussian noise to synthesize clean latents, which are decoded back to pixel space.

Event Representation. Event cameras capture asynchronous streams $\{e_i\}_{i=1}^N$ where $e_i = (x_i, y_i, t_i, p_i)$. To enable frame-based processing, we discretize these events into a voxel grid $V \in \mathbb{R}^{B\times H\times W}$ [Ercan et al. 2024] by accumulating polarity $p_i \in \{\pm1\}$ into B temporal bins via linear interpolation:

$$V(t, y, x) = \sum_i p_i \max(0, 1 - |t - t_i^*|)\delta(x - x_i, y - y_i), \quad (1)$$

where δ is the Kronecker delta and $t_i^* \in [0, B-1]$ represents the normalized timestamp relative to duration ΔT. In our experiments, we set $B = 3$.

3.2 Event-based Video Generation Framework

We denote the current video frames as $X = \{x_0, \ldots, x_{F-1}\}$. To align asynchronous events with discrete frames, we partition the event stream into time windows. Specifically, the event voxel v_k corresponds to frame x_k at timestamp t_k by aggregating events within the interval $[t_{k-1}, t_k)$ via the transformation in Sec. 3.1. This yields a synchronized event sequence $V = \{v_1, \ldots, v_{F-1}\}$.

Context and Autoregressive Unrolling. Naive long video generation often uses the last frame of the previous chunk as the initial condition for the next. However, this recursive process causes error accumulation and artifact propagation. To mitigate this, we employ context frames and context event voxels as history conditions. Specifically, context frames are defined as the continuous sequence immediately preceding the first frame of the current chunk, and context event voxels are the corresponding event voxels associated with these frames.

However, simply conditioning on history introduces a domain gap between training and inference, as the model conditions on Ground Truth context frames during training but relies on its own predictions during inference. To bridge this gap, we propose an

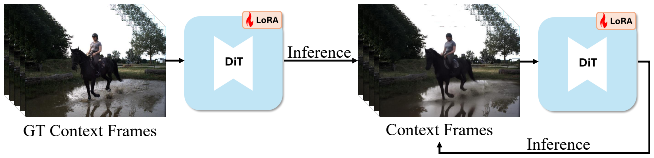

Fig. 3. **Autoregressive Unrolling.** To bridge the domain gap between training and inference, we employ an iterative training strategy. Initially, the model is trained with Ground Truth (GT) context frames for convergence (left). Subsequently, we activate the unrolling mechanism by performing an inference pass to generate predictions, which then replace the GT context frames for fine-tuning (right). This iterative feedback loop forces the model to adapt to its own generation errors, mitigating error accumulation during long video generation.

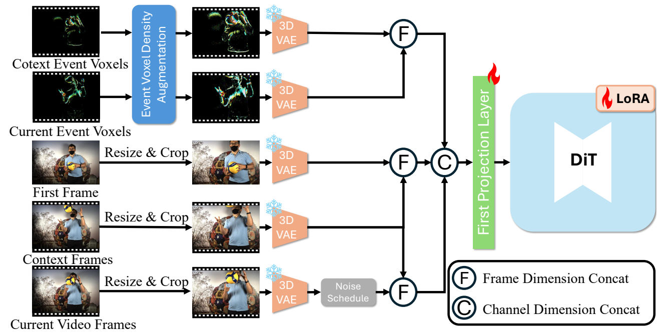

Fig. 4. **Overview of our training pipeline.** To enhance robustness against sensor variations, input event voxels undergo Event Voxel Density Augmentation, and the first frame, context frames, and current video frames are synchronously resized and cropped to ensure spatial alignment. All inputs are encoded into latents via a frozen 3D VAE. These latents are aligned and fused through frame dimension concatenation and channel dimension concatenation. Finally, we expand and fully fine-tune the First Projection Layer to accommodate the additional event channels, while the DiT backbone is efficiently fine-tuned using LoRA.

Autoregressive Unrolling training strategy. Initially, the model is trained with GT context frames until convergence. We then activate the unrolling mechanism where predictions generated on the training set are substituted as context frames for subsequent fine-tuning, as illustrated in Fig. 3. This iterative unrolling is repeated T times, forcing the model to adapt to its own prediction errors and effectively aligning the training distribution with inference behavior.

Event Voxel Density Augmentation. The spatial density of event voxels varies significantly due to diverse sensor resolutions and scene depths. To enhance robustness, we introduce a density augmentation strategy during training. Specifically, we randomly resize event voxels while preserving the aspect ratio, bounded within a dynamic range $[S_{min}, S_{max}]$. The lower bound S_{min} is set slightly larger than the network input to facilitate random cropping, while S_{max} is capped at 2× the original resolution. Following FireNet [Scheerlinck et al. 2020], we normalize based on the statistics of non-zero values to preserve sparsity. Finally, a random crop is applied to match the network input resolution. To maintain spatial alignment, the identical geometric transformations are synchronously applied to the first frame, context frames, and current video frames.

Finetuning with Event Voxels and Context. Following spatial alignment, we ensure temporal alignment of the input data prior to fine-tuning. Since the first frame x_0 is provided, we zero-pad the current event voxels V at the initial timestamp to align with the subsequent frames $x_{1:F-1}$. All inputs, including x_0, padded V, and context elements, are encoded via a frozen 3D VAE. Note that constructing V with $B = 3$ allows the event stream to be processed natively by the VAE's standard 3-channel input. We then construct the final input latents by concatenating three temporally-aligned sequences along the channel dimension: (1) context latents with the zero-padded first-frame latent Z_{x_0}, (2) context latents with the noise latents Z_t, and (3) context event latents with the current event latents.

To accommodate the additional event condition, we augment the first projection layer within the patchify module, extending the weights $\mathbf{W}_{in} \in \mathbb{R}^{D\times 2C_{vae}\times K\times K}$ to $\mathbf{W}^*_{in} \in \mathbb{R}^{D\times 3C_{vae}\times K\times K}$. During training, we fully fine-tune this expanded layer and employ LoRA for the DiT blocks, as shown in Fig. 4. The loss is calculated exclusively on the Z_t component. Additionally, we apply a 5% dropout to Z_{x_0} to enhance reconstruction robustness and introduce text prompts with a 20% probability for text-guided generation.

3.3 Event-based Video Reconstruction, Prediction, and Frame Interpolation.

We formulate event-based video reconstruction, prediction, and frame interpolation as tasks of event-based video generation. By adjusting the input conditions, the model adapts to specific task requirements:

- **Video Reconstruction:** The model recovers photometric details solely from events. For the first chunk, start frame and context are empty, forcing the model to reconstruct the scene from scratch. (see Sec. 4.1 for details) For subsequent chunks, the last frame of the preceding chunk serves as the first frame, seamlessly transitioning the task into a prediction paradigm.
- **Video Prediction:** Given the start frame, the event stream, and context, the model generates subsequent video frames by leveraging the start frame for initial texture guidance, the event stream for dynamic information, and the context for historical reference. For the first chunk, the context frames are populated by replicating the start frame to fill the temporal window, while context event voxels are initialized as empty (see Sec. 4.1 for details).
- **Video Frame Interpolation:** Given the start frame and end frame as boundary conditions, the model synthesizes intermediate frames by leveraging both forward and backward event streams. These streams act as a dense motion guide bridging the boundaries.

Adaptive Context Switch. Standard autoregressive updates after every chunk cause error accumulation. To mitigate this, we introduce an Adaptive Context Switch. While the first chunk requires a mandatory update, subsequent updates are dynamic based on context relevance. We re-input the denoised latents $\hat{Z}_0$ into the DiT to extract the attention map $\mathbf{A}$ and compute the average attention weight μ_{attn} of current tokens attending to context tokens:

$$\mu_{attn} = \frac{1}{L \times H \times N_{curr}} \sum_{l=1}^{L} \sum_{h=1}^{H} \sum_{i \in \text{Current}} \sum_{j \in \text{Context}} \mathrm{A}_{i,j}^{(l,h)}, \quad (2)$$

where L is the number of layers, H is the number of Attention Heads, and N_{curr} is the number of Current Tokens. If $\mu_{attn} \geq \tau$, the existing context is retained to prevent drift. If $\mu_{attn} < \tau$, indicating low relevance, we trigger a Context Switch with a single-attempt retry mechanism: the current generation is discarded, the context is updated to the immediate predecessor, and the chunk is regenerated. In our experiments, we set $\tau = 0.05$.

Reencoding Alignment. We extend Time Reversal Fusion [Feng et al. 2024] for zero-shot interpolation. Standard methods directly flip backward latents, but due to temporal compression in 3D VAEs [Wan et al. 2025; Yang et al. 2024], latent and pixel space operations are not commutative:

$$\mathrm{Flip}_{lat}(Z_t) \neq \mathcal{E}(\mathrm{Flip}_{pix}(\mathcal{D}(Z_t))), \quad (3)$$

where $\mathcal{D}, \mathcal{E}$ are the decoder/encoder and $\mathrm{Flip}_{lat/pix}$ denote temporal flipping. This discrepancy causes misalignment. We propose Reencoding Alignment to rectify this by decoding the predicted clean latents $\hat{Z}_0$, flipping in pixel space, and re-encoding:

$$\tilde{Z}_0^{fwd} = \mathcal{E}(\mathrm{Flip}_{pix}(\mathcal{D}(\hat{Z}_0^{fwd}))), \quad \tilde{Z}_0^{bwd} = \mathcal{E}(\mathrm{Flip}_{pix}(\mathcal{D}(\hat{Z}_0^{bwd}))). \quad (4)$$

This ensures precise temporal alignment between the bidirectional branches (Fig. 5).

Cross Residual Correction. Although Reencoding Alignment resolves misalignment, the extra Decode-Encode loop induces reconstruction loss. To compensate, we introduce Cross Residual Correction, inspired by LookingGlass [Chang et al. 2025]. We design a Cross Injection strategy leveraging complementary details from forward and backward branches to mutually restore information. We compute the residual between the original $\hat{Z}_0$ and re-encoded $\bar{Z}_0 = \mathcal{E}(\mathcal{D}(\hat{Z}_0))$ latents:

$$\Delta\hat{Z}_0^{fwd} = \hat{Z}_0^{fwd} - \bar{Z}_0^{fwd}, \quad \Delta\hat{Z}_0^{bwd} = \hat{Z}_0^{bwd} - \bar{Z}_0^{bwd}. \quad (5)$$

We then inject the opposing residual into the aligned latents:

$$Z_0^{\prime bwd} = \tilde{Z}_0^{bwd} + \Delta\hat{Z}_0^{fwd}, \quad Z_0^{\prime fwd} = \tilde{Z}_0^{fwd} + \Delta\hat{Z}_0^{bwd}. \quad (6)$$

This recovers high-frequency details and promotes Temporal Consensus (Fig. 5). Finally, the corrected latents are fused via alpha blending and re-noised for the subsequent loop.

4 Experiments

4.1 Experiment Settings

Dataset. We train exclusively on the BS-ERGB [Tulyakov et al. 2021] training set, chosen for its high-quality real-world events and large motion, after filtering sequences with missing data. For reconstruction and prediction, we follow the EVREAL benchmark [Ercan et al. 2023] on selected subsets of three real-world event datasets: ECD [Mueggler et al. 2017], MVSEC [Zhu et al. 2018], and HQF [Stoffregen et al. 2020]. Crucially, these sequences cover a wide range of temporal durations, spanning from short clips in ECD (~300 frames) to long-term sequences in MVSEC and HQF (up to 2,740 and 2,430 frames, respectively), evaluating the model's stability over extended periods. For frame interpolation, we evaluate on the BS-ERGB test set and HQF dataset.

Implementation Details. We employ CogVideoX I2V as our backbone, generating 49-frame chunks at 720×480. In training, we apply LoRA ($r = 64$) to the DiT blocks and full finetune the first projection layer. Training involves 3 times autoregressive unrolling. A 20-frame context is maintained during both training and inference. Specifically, for the first chunk during inference where historical context is unavailable, we apply task-specific initialization: reconstruction employs zero tensors for both the start frame latent and context latents; prediction replicates the start frame $20\times$ to populate the context; and frame interpolation replicates the start and end frames $10\times$ each to form the context. Across all tasks, the corresponding context event voxels are consistently zero-padded. During evaluation, inspired by VDM-EVFI, we upsample inputs to mitigate detail loss from VAE encoding. Specifically, inputs below the model's input resolution are upsampled to match it, whereas those exceeding it are upsampled by $\approx 2\times$ and processed using their proposed Per-tile Denoising and Fusion strategy.

4.2 Comparisons with SOTA Event base Reconstruction and Prediction Methods

For reconstruction, we benchmark against SOTA E2V methods, including E2VID, FireNet, E2VID+, FireNet+, SPADE-E2VID, SSL-E2VID, ET-Net, and HyperE2VID, using pre-trained weights from EVREAL. For prediction, we compare with VDM-EVFI using its official 13-frame pre-trained weight in the forward generation setting. Since event streams lack absolute intensity information, for a fair comparison, we align the global brightness of all generated results with the ground truth sequences.

As shown in Fig. 6, the 2th row illustrates the later stages of the sequences, where existing SOTA models suffer from severe error accumulation in both tasks. The 3th row depicts the early stages; here, SOTA E2V models struggle to reconstruct high-quality images due to the initialization limitations of their recurrent architectures. Notably, VDM-EVFI exhibits noticeable error accumulation even in the early part of the sequence. Both Tab. 1 and Fig. 6 demonstrate that our method consistently outperforms baselines across all three datasets, validating its superiority in handling both short and long-term sequences. More visual comparisons are shown in Fig. 11.

4.3 Comparisons with SOTA Event base Video Frame Interpolation Methods

Although our model was trained exclusively for reconstruction and prediction, we extended it to Event-based Video Frame Interpolation (EVFI) without any fine-tuning. By utilizing the identical weights, we evaluated our method in a zero-shot setting against dedicated SOTA EVFI baselines (CBMNet-Large [Kim et al. 2023] and TLXNet+ [Ma et al. 2024a]) on interpolating 31 frames.

Visual comparisons reveal significant limitations in the baselines when handling large-frame interpolation on real-world data. As shown in Row 1 of Fig. 7, CBMNet-Large suffers from severe ghosting and color artifacts in dynamic human motions, while TLXNet+ fails to preserve structural integrity. Furthermore, in Rows 2, our

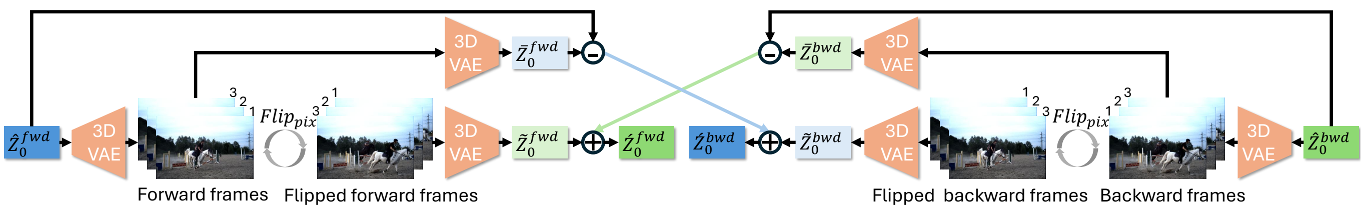

Fig. 5. **Reencoding Alignment and Cross Residual Correction.** To address temporal misalignment caused by the discrepancy between latent-space and pixel-space flipping, we propose Reencoding Alignment. The denoised latents, $\hat{Z}_0^{fwd}$ and $\hat{Z}_0^{bwd}$, are decoded into pixel space, flipped temporally ($Flip_{pix}$), and then re-encoded via the 3D VAE to yield the aligned latents $\tilde{Z}_0^{fwd}$ and $\tilde{Z}_0^{bwd}$. To mitigate information loss inherent in this re-encoding process, we employ Cross Residual Correction. We calculate the residual difference between the original and the re-encoded latents (e.g., the subtraction node $\hat{Z}_0^{fwd} - \bar{Z}_0^{fwd}$) and inject this detail information into the opposite branch. Specifically, the forward residual is added to the backward aligned latents $\tilde{Z}_0^{bwd}$ to produce the final corrected latents $Z_0'^{bwd}$, and symmetrically, the backward residual is injected into $\tilde{Z}_0^{fwd}$ to obtain $Z_0'^{fwd}$. This symmetric Cross Injection mechanism promotes temporal consensus between branches while preserving fine-grained details. Light-colored boxes represent information loss.

Table 1. **Quantitative results on ECD [Mueggler et al. 2017], MVSEC [Zhu et al. 2018], and HQF [Stoffregen et al. 2020].** Comparing our method against SOTA baselines (**Red**: best; blue: second). In *Reconstruction*, we consistently achieve the best LPIPS scores, indicating superior perceptual quality compared to regression-based methods. In *Prediction*, our method significantly outperforms VDM-EVFI across all metrics and datasets, validating our robust long-term generation capabilities.

		ECD [Mueggler et al. 2017]			MVSEC [Zhu et al. 2018]			HQF [Stoffregen et al. 2020]		
Method	Venue	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Reconstruction										
E2VID [Rebecq et al. 2019b]	TPAMI 2019	**22.40**	0.690	0.177	14.84	0.354	0.576	15.06	0.522	0.362
FireNet [Scheerlinck et al. 2020]	WACV 2020	20.01	0.597	0.241	15.51	0.374	0.587	13.89	0.445	0.486
E2VID+ [Stoffregen et al. 2020]	ECCV 2020	22.32	0.676	0.147	16.27	0.411	0.463	15.97	0.552	0.256
FireNet+ [Stoffregen et al. 2020]	ECCV 2020	21.18	0.621	0.215	15.45	0.389	0.498	15.64	0.491	0.314
SPADE-E2VID [Cadena et al. 2021]	TIP 2021	19.71	0.569	0.301	15.70	0.389	0.538	13.36	0.417	0.538
SSL-E2VID [Paredes-Vallés and De Croon 2021]	CVPR 2021	20.00	0.615	0.278	14.36	0.353	0.633	13.51	0.446	0.474
ET-Net [Weng et al. 2021]	ICCV 2021	22.26	0.675	0.159	15.97	0.409	0.435	16.17	0.542	0.263
HyperE2VID [Ercan et al. 2024]	TIP 2024	**22.40**	0.685	0.157	16.13	0.417	0.434	16.42	0.539	0.257
Ours	-	22.28	**0.708**	**0.139**	**16.83**	**0.440**	**0.405**	**16.45**	**0.603**	**0.240**
Prediction (w/ start frame)										
VDM-EVFI [Chen et al. 2025a]	CVPR 2025	20.33	0.614	0.244	15.46	0.276	0.668	13.52	0.373	0.520
Ours	-	**24.40**	**0.771**	**0.110**	**18.18**	**0.502**	**0.359**	**16.67**	**0.619**	**0.229**

Table 2. **Interpolation (31 skips) on ERGB [Tulyakov et al. 2021] and HQF [Stoffregen et al. 2020].** Comparing our *zero-shot* method vs. *supervised* baselines (**Red**: best; blue: second), we excel in *LPIPS* and generalization. Unlike regression baselines, which blur textures to boost PSNR, our generative approach preserves high-frequency details.

	Training	BS-ERGB			HQF		
Method	setting	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
CBMNet-Large	Supervised	**24.59**	**0.767**	0.170	24.55	0.767	0.166
TLXNet+	Supervised	18.85	0.641	0.226	17.18	0.484	0.352
Ours	Zero-shot	24.40	0.744	**0.124**	**25.39**	**0.800**	**0.105**

method is the only one capable of reconstructing clear, readable text, whereas CBMNet-Large results in severe blur, and TLXNet+ produces misaligned, ghosted outputs. Both Tab. 2 and Fig. 7 confirm that our zero-shot approach surpasses existing SOTA methods designed specifically for EVFI. More visual comparisons are shown in Fig. 12.

4.4 Ablation Study

Ablation Study on Event-based Video Reconstruction. We analyze component effectiveness in Fig. 8 and Tab. 3. First, the pretrained prior is critical (Row 1); training the diffusion backbone from scratch on limited data fails to converge, yielding pure noise. Second, relying solely on event voxels without context mechanisms causes severe error accumulation and artifacts (Row 2). Third, removing Autoregressive Unrolling and Adaptive Context Switch (Row 3) exposes a training-inference domain gap; training with Ground Truth context while inferring with predictions causes drift, evidenced by point artifacts. Fourth, ablating only the Adaptive Context Switch (Row 4) by forcing updates after every chunk results in error accumulation, as shown by grid artifacts. These results confirm that each component is vital for minimizing error accumulation and maintaining stability.

Ablation Study on Event-based Video Frame Interpolation. We validate our method by ablating Reencoding Alignment, Cross Residual Correction, and Event Voxel Data Augmentation (Tab. 4, Fig. 9).

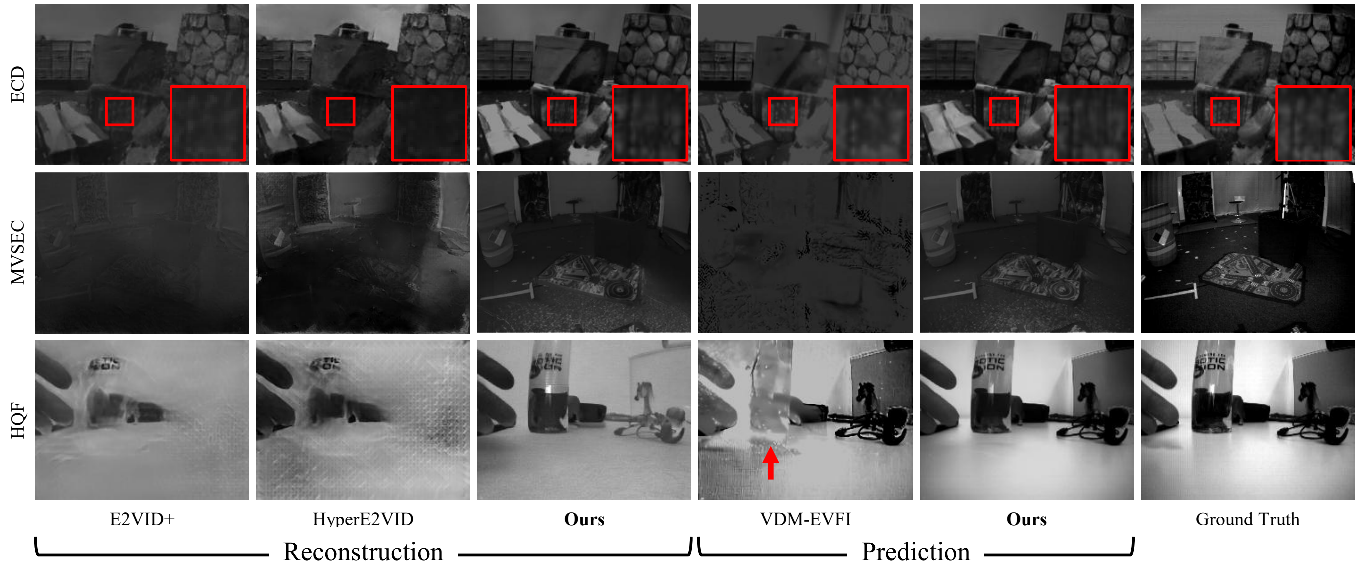

Fig. 6. **Qualitative comparisons on ECD [Mueggler et al. 2017], MVSEC [Zhu et al. 2018], and HQF [Stoffregen et al. 2020] datasets.** Our LongE2V recovers high-frequency textures where regression baselines (E2VID+, HyperE2VID) suffer from blurring (Row 1). In prediction tasks, we avoid the severe noise accumulation and ghosting artifacts (red arrow) seen in VDM-EVFI (Rows 2–3), maintaining superior structural fidelity and temporal stability.

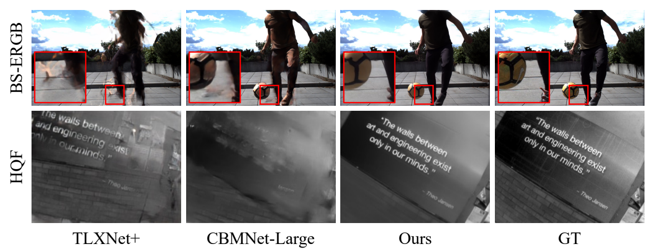

Fig. 7. **Zero-shot interpolation on BS-ERGB and HQF.** Baselines (TLXNet+, CBMNet-Large) suffer from structural collapse or blur under large motion (*Top*), whereas our LongE2V captures accurate dynamics. On fine text (*Bottom*), our Reencoding Alignment eliminates the ghosting seen in baselines, ensuring legibility.

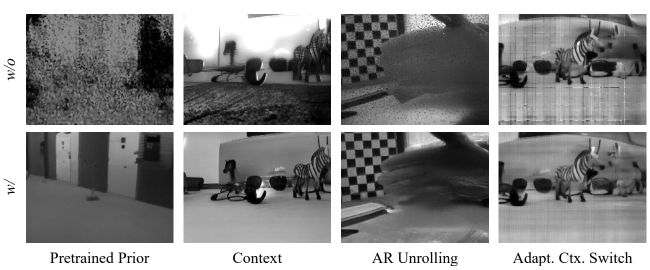

Fig. 8. **Qualitative ablation on reconstruction.** *Top*: Ablated variants; *Bottom*: Full Method. w/o Pretrained prior yields noise; w/o Context causes structural ambiguity; w/o Autoregressive unrolling leads to point artifacts (drift); and w/o Adaptive context switch causes grid artifacts. Our method ensures stable, high-fidelity results.

Table 3. **Ablation of reconstruction components on HQF dataset.** The *pretrained prior* is essential for convergence (Row 1 vs. 2). Including *context*, *Autoregressive unrolling*, *Adaptive Context Switch* effectively mitigates long-term drift (Rows 3, 4, and 5), allowing the full method to achieve the best performance across all metrics.

Pretrained prior	Context	AR unroll.	Adaptive ctx. switch	PSNR↑	SSIM↑	LPIPS↓
	✓	✓	✓	10.26	0.094	0.746
✓				13.81	0.508	0.320
✓	✓			15.63	0.561	0.278
✓	✓	✓		16.42	0.595	0.247
✓	✓	✓	✓	**16.45**	**0.603**	**0.240**

Without Reencoding Alignment, directly flipping latents causes temporal misalignment, resulting in significantly blurred dynamic figures. This confirms the necessity of our decode-flip-encode process for spatial coherence. Removing Cross Residual Correction leads to ghosting and semi-transparent artifacts due to 3D VAE information loss; restoring this via cross-injection is crucial for Temporal Consensus, improving LPIPS by 0.037. Finally, omitting Event Voxel Data Augmentation exposes the model to a density mismatch between training and inference, where upsampling the input resolution at test time dilutes event density per voxel, leading to unstable generation with color deviations and physical artifacts such as black stripes on the basketball.

4.5 Text-Guided Event Video Colorization

Leveraging the inherent text-guided generation capabilities of our backbone, CogVideoX I2V, we incorporate text prompts during training with a 20% probability. This strategy enables the model to adapt

Fig. 9. **Visual ablation on interpolation.** *w/o Reencoding Alignment* causes ghosting due to latents misalignment; *w/o Cross Residual Correction* blurs fine details due to VAE loss; and *w/o Event Voxel Density Augmentation* yields artifacts from density mismatch. Our *Full Method* restores sharp, coherent details comparable to *Ground Truth*.

Table 4. **Ablation of interpolation components on BS-ERGB dataset.** Reencoding Alignment is critical for structural fidelity, while Cross Residual Correction enhances perceptual quality (LPIPS). Event Voxel Density Augmentation improves robustness, with the full method achieving superior results.

Reencoding alignment	Cross res. correction	Event voxel density aug.	PSNR↑	SSIM↑	LPIPS↓
		✓	20.58	0.670	0.202
✓		✓	24.14	0.734	0.161
✓	✓		22.58	0.728	0.129
✓	✓	✓	**24.40**	**0.744**	**0.124**

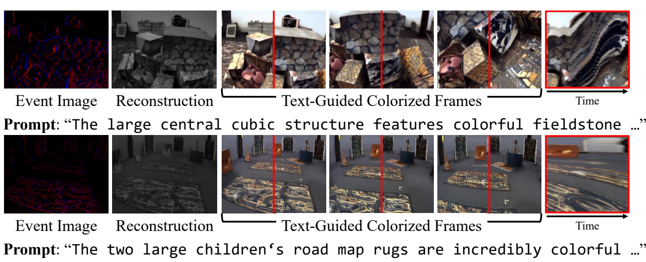

Fig. 10. **Text-Guided Event Video Colorization.** *Left to Right*: Input events; standard reconstruction (geometry baseline); text-stylized frames; and a temporal XT-slice (at **red line**). The results demonstrate our model effectively decouples event-driven motion from text-defined appearance, while the XT-slice confirms the generated textures are temporally coherent.

to a multi-modal conditioning setting, learning to synthesize videos based on both text prompts and event voxels. As demonstrated in Fig. 10, when provided with only event voxels and a text prompt, the model successfully reconstructs the structural textures of the scene from the event data while applying coloration and stylization based on the textual description, effectively achieving event video colorization.

5 Conclusion

We presented LongE2V, leveraging video diffusion priors for event-based reconstruction, prediction, and frame interpolation. We mitigate temporal drift via Autoregressive Unrolling and Adaptive Context Switching, while ensuring interpolation consistency using Reencoding Alignment and Cross Residual Correction. Experiments confirm LongE2V outperforms state-of-the-art methods in perceptual quality and robustness. Future work will focus on accelerating inference and exploring efficient memory mechanisms for long-term consistency.

Acknowledgments

This research was funded by the National Science and Technology Council, Taiwan, under Grants NSTC 112-2222-E-A49-004-MY2 and 113-2628-E-A49-023-. The authors are grateful to Google, NVIDIA, and MediaTek Inc. for their generous donations. Yu-Lun Liu acknowledges the Yushan Young Fellow Program by the MOE in Taiwan.

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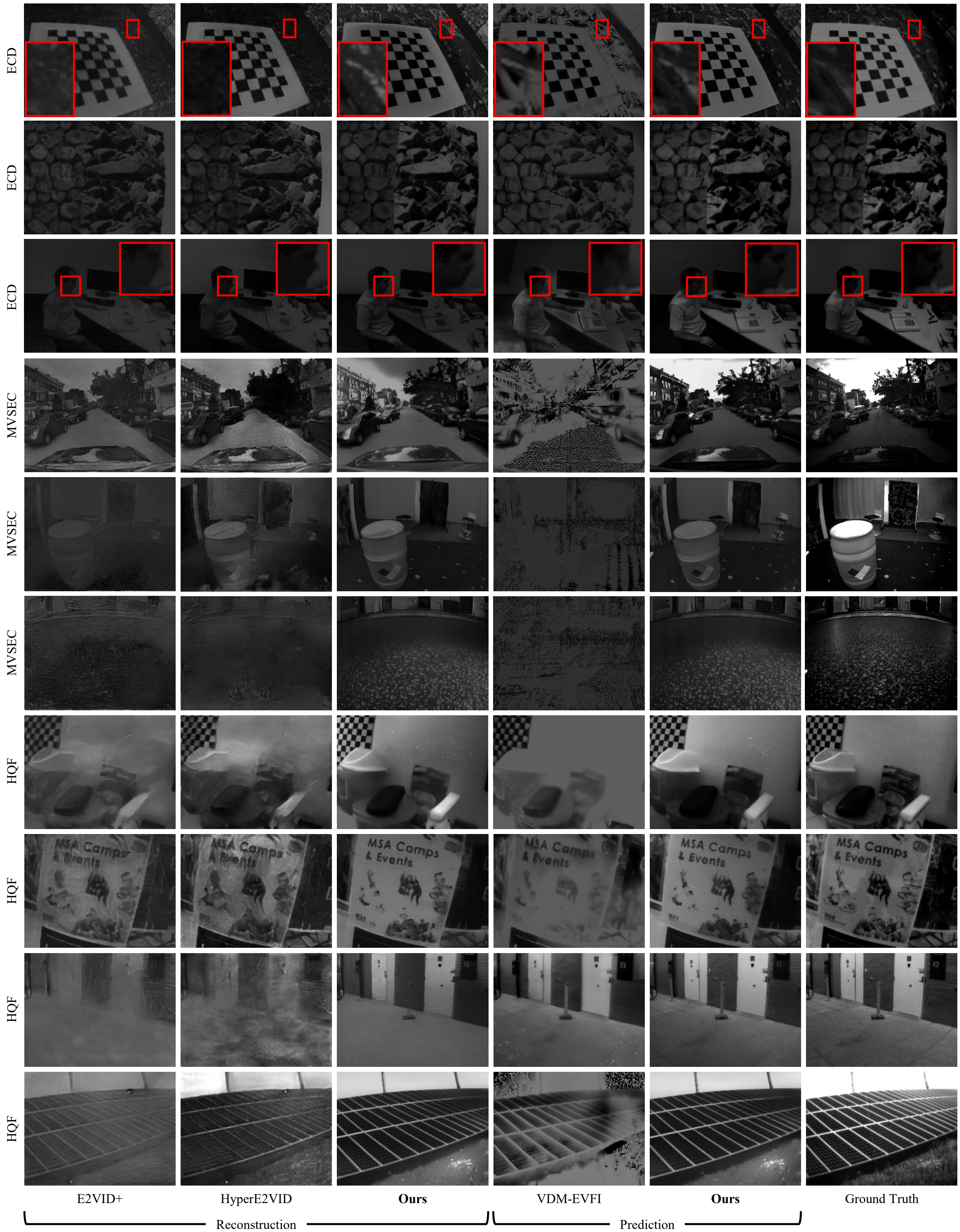

Fig. 11. **Additional qualitative comparisons on ECD [Mueggler et al. 2017], MVSEC [Zhu et al. 2018], and HQF datasets [Stoffregen et al. 2020].**

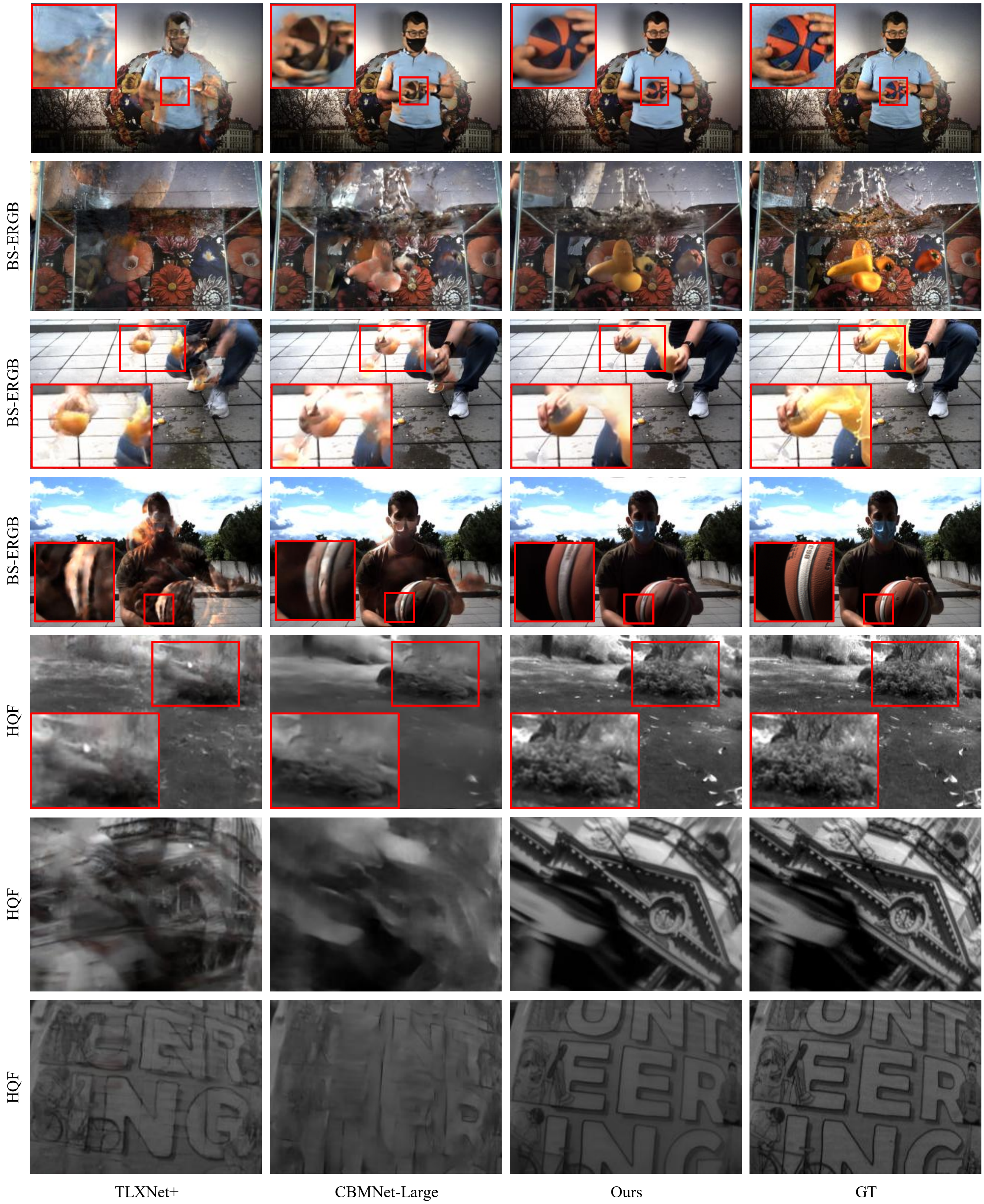

Fig. 12. **Additional zero-shot interpolation results on BS-ERGB [Tulyakov et al. 2021] and HQF [Stoffregen et al. 2020] datasets.**